EFY TESTED Ashwini K. Sinha

TOUCHLESS DOORBELL THAT RINGS
Only When Hand Is Sanitised

RAKESH JAIN AND RAHUL MOUD

Touching of a doorbell switch by different people can spread coronavirus and many other communicable diseases from one person to the others. So, protection from such diseases at the entrance doorbell switch is as much necessary as the use of a hand sanitiser. This project is meant to help in combating the spread of viruses and diseases that might transfer during the touching of doorbells.

This touchless doorbell can be made easily with the help of an Arduino development board. The bell rings only when the visitor's hands are found to be sanitised. To ring the bell, the visitor needs to bring her/his hands just 3cm to 5cm away from the MQ3 gas sensor. Author's

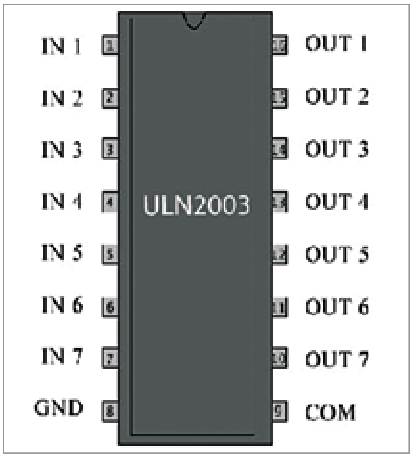

Fig. 3: IC ULN2030 pin details

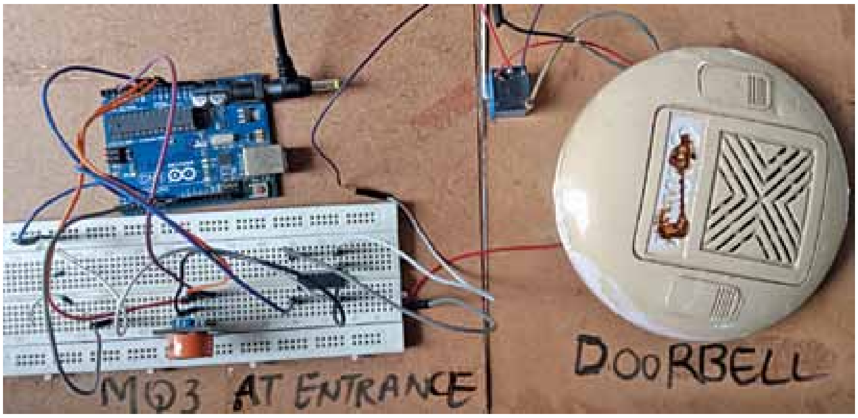

Fig. 1: Author's prototype of the bell

prototype is shown in Fig. 1.

Circuit diagram of the touchless doorbell is shown in Fig. 2. The circuit comprises an Arduino board, gas sensor MQ3 (which can also sense alcohol fumes in an alcohol based sanitiser), relay driver ULN20003 IC, 5V single changeover relay RL1, and a few other components.

When someone brings a hand near the MQ3 sensor to ring doorbell, if it senses alcohol vapours arising from the sanitised hand, it generates an analogue signal. This signal drives the interfaced Arduino-Uno to generate signals at pin 8 and pin 9. The LED connected to pin 8 glows and pin 9 connected to ULN2003 IC drives

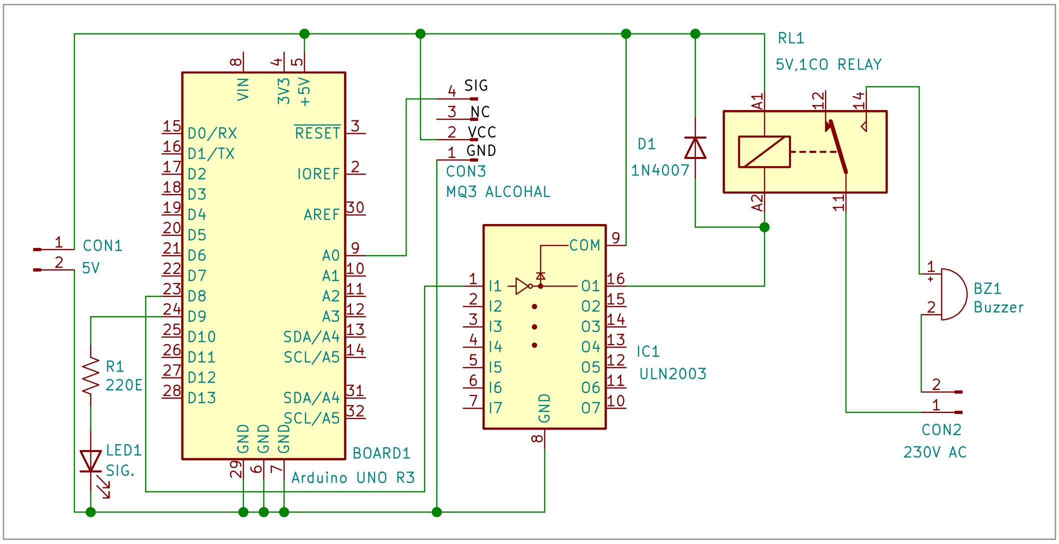

Fig. 2: Circuit diagram of the bell